

Factors Impacting Income Using Census Data

Review of Requirements

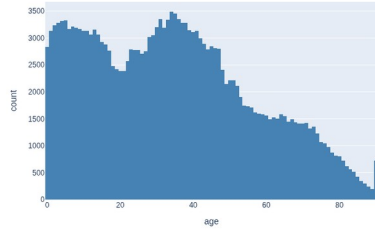
- Identify metrics that associated with a person making more or less than \$50,000 per year
- Deliver a pipeline that can take in input of data and export the prediction for future use.
- The goal is to predict need of social services in an area without using any financial info
- Leave in children, unemployed and retired in data

Steps

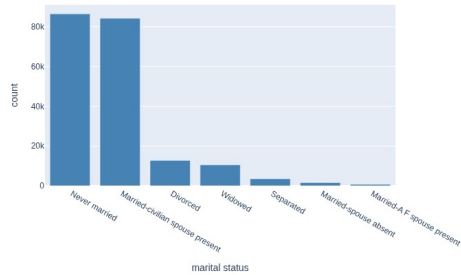
- Data Exploration
- Prep Data/Feature Engineering
- Selecting and Training Model
- Analyze Results

Data Exploration

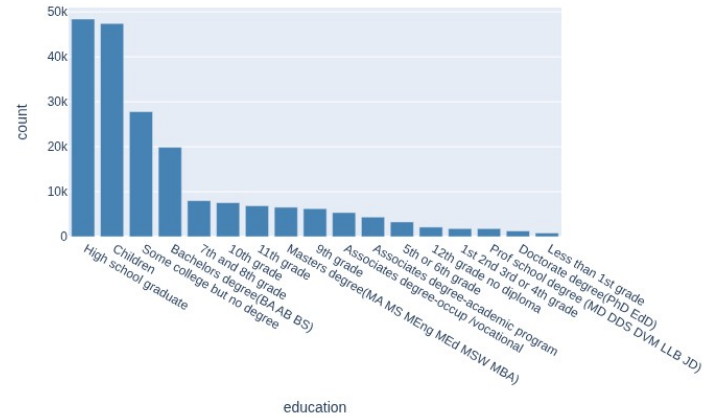
age distribution



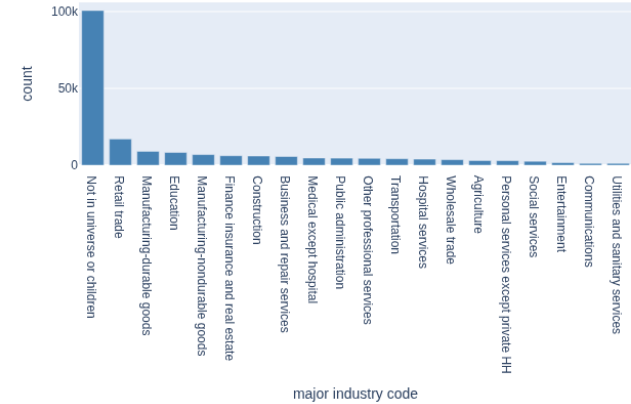
marital status value counts



education value counts

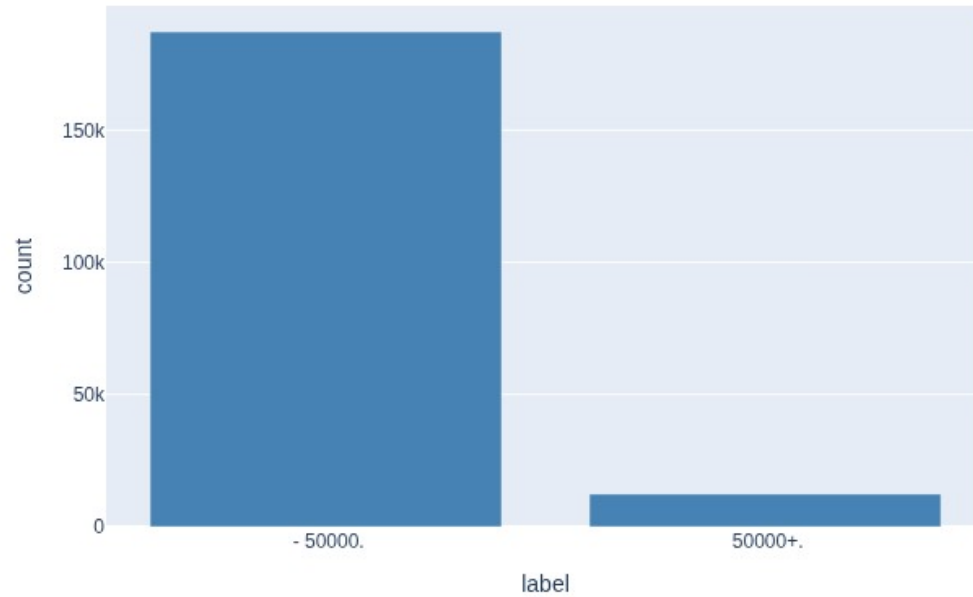


major industry code value counts



+/- 50k

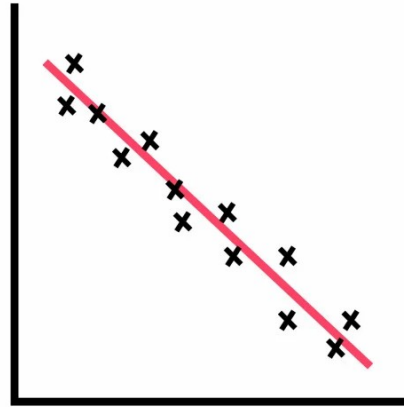
label value counts



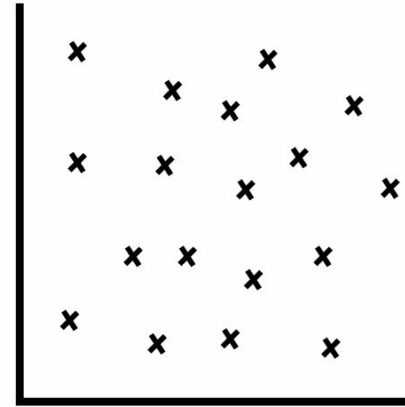
Data Correlations



Positive
Correlation

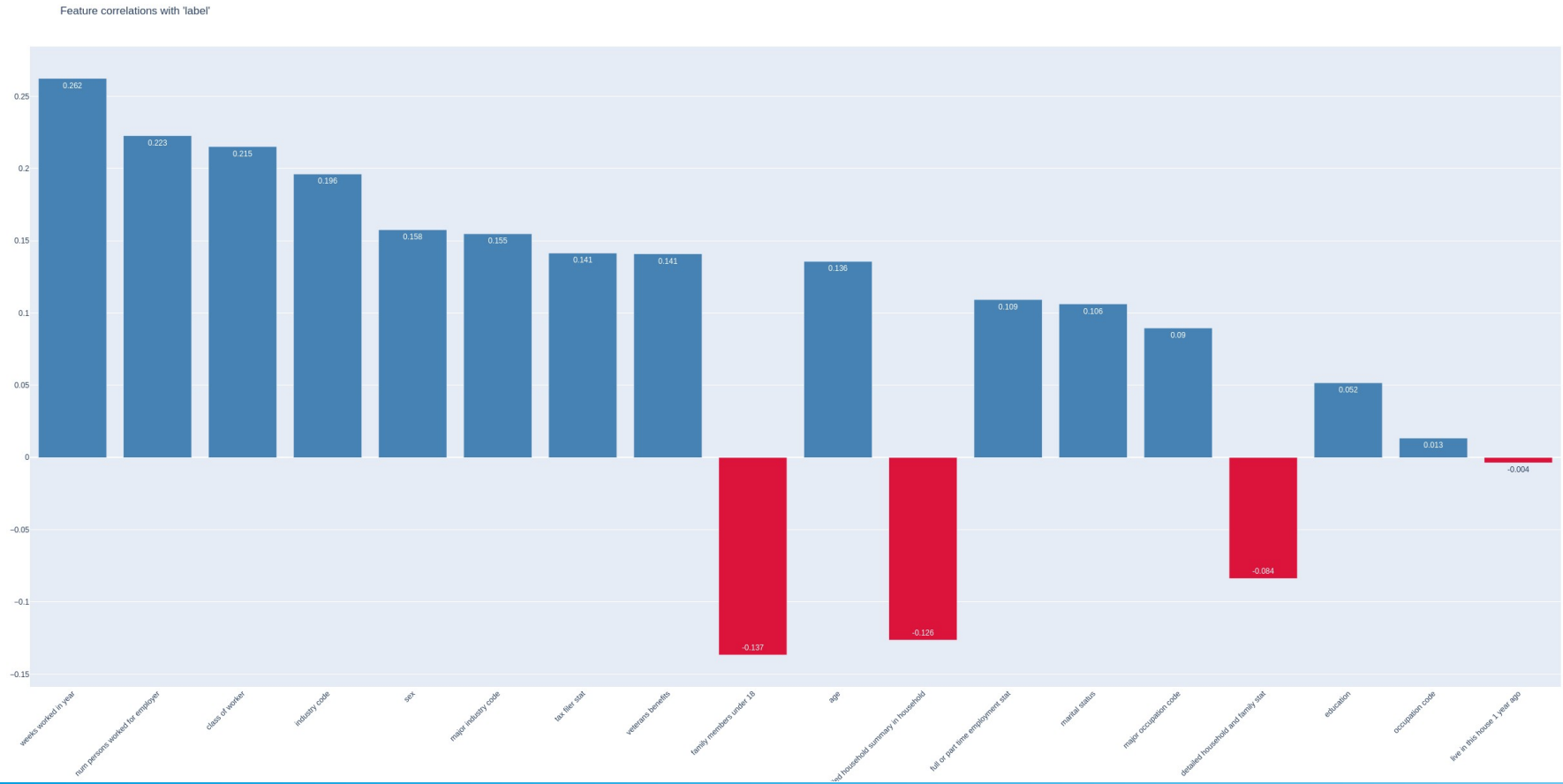


Negative
Correlation



No
Correlation

Data Correlations

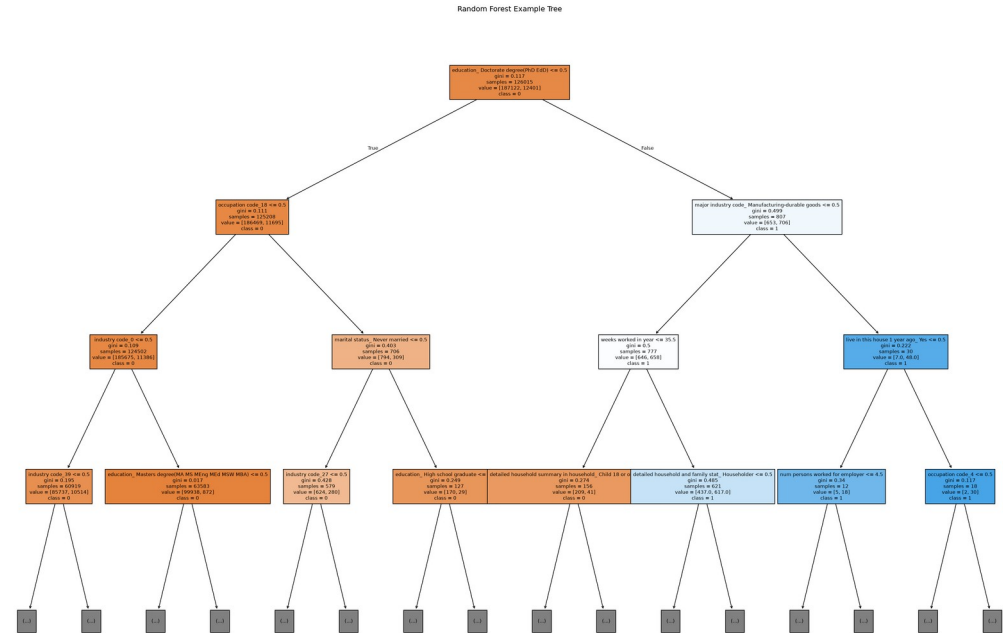


Data Prep

- Data received was skewed to most making under 50k
 - Removing variables that had little impact on outcome
 - One Hot Encoding – transform string(text) data into numbers for the model
- Data Dropped
 - total person income – the label that we are trying to predict is derived from this field
 - instance weight – the sampling rate, shouldn't be used for classifiers
 - Other Low impact features
 - Other Financial Data

First Pass Model

- Random Forest is the clear first choice here
- This is a great model
 - Easy to train
 - Easy to understand what the model is doing
- 95% Accuracy
 - But is more accurate one way vs the other



Classification report:

	precision	recall	f1-score	support
0	0.96	0.98	0.97	93576
1	0.60	0.36	0.45	6186
accuracy			0.95	99762
macro avg	0.78	0.67	0.71	99762
weighted avg	0.94	0.95	0.94	99762

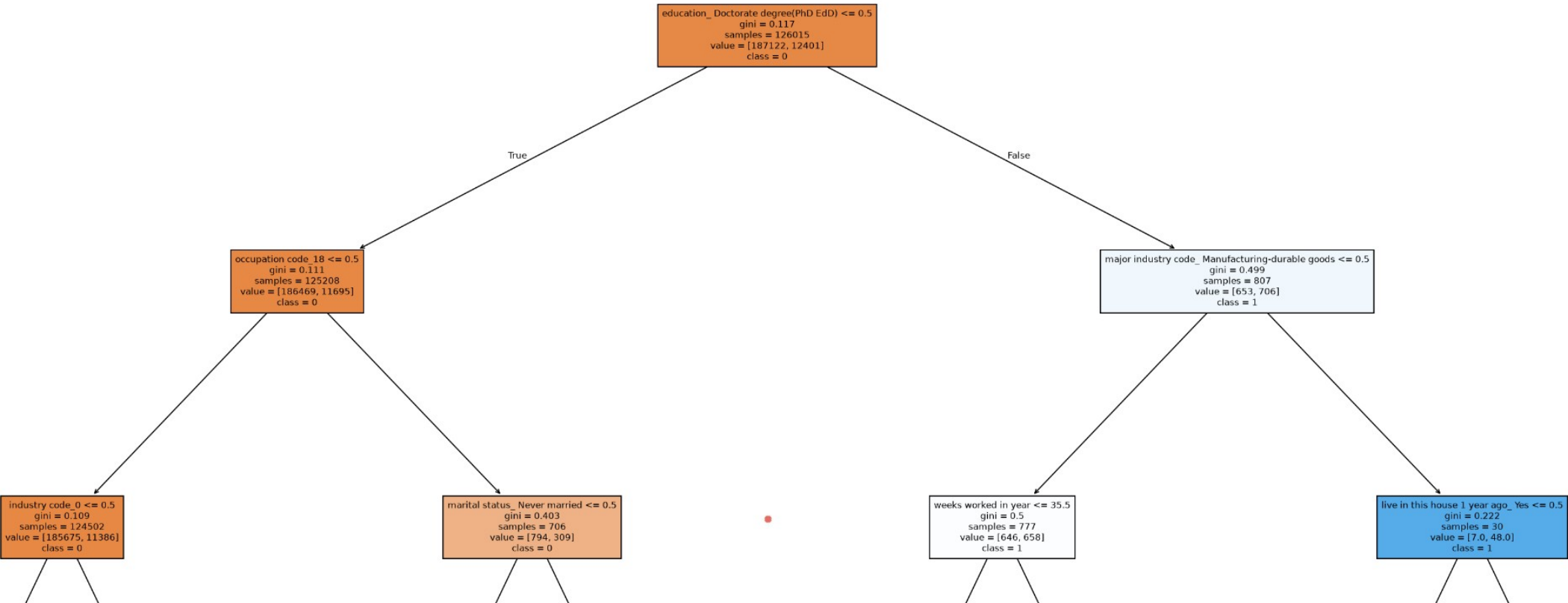
model predicted as positive, how many

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

- **Recall:** The model's ability to find all p examples in the data, how many did th

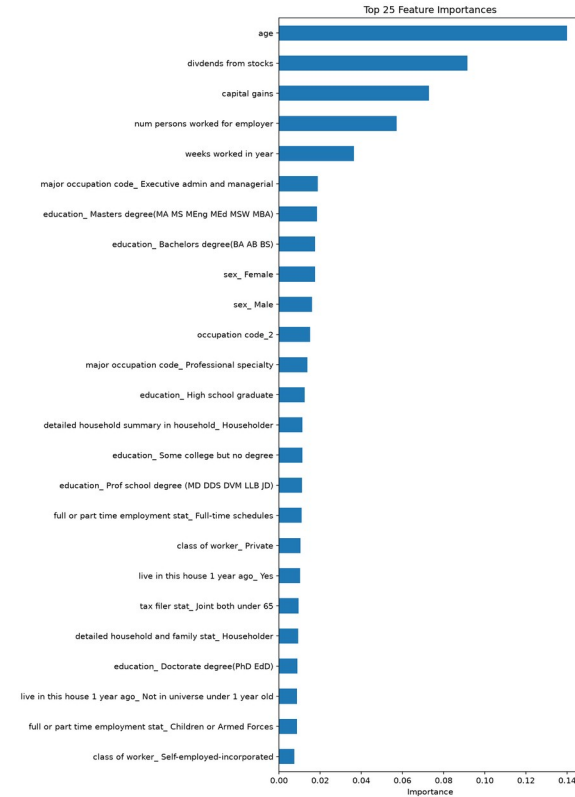
$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

First Pass Model



Feature Importance

- Age
- Dividends from Stocks and Capital Gains
 - This was removed from the final model do one making more or less than 50k can be derived from this
- Size of Company
- Weeks Work in a year
- Occupation and Education

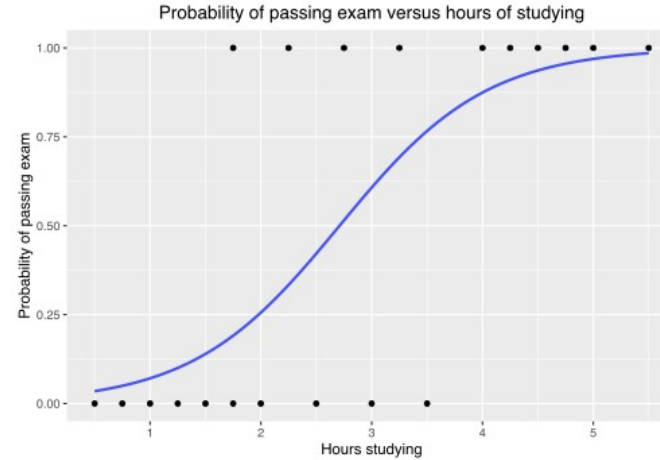
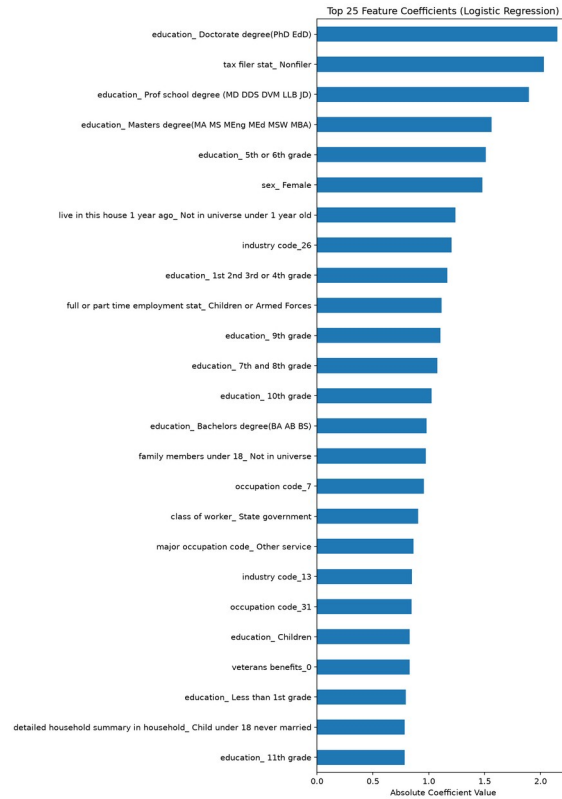


Hyper Parameter Tuning

- Random Forest and other models have different settings you can tune. Hyper Parameter tuning tries to auto find the best settings for the model.
- We didn't get a mass improvement from this but we can see the best settings for Random Forest are
 - Estimators: 400
 - Min Sample Split 5
 - Min Samples Leaf 2
 - Max Features: sqrt
 - Max Depth: None

params	split0_test_score	split1_test_score	split2_test_score	mean_test_score	std_test_score	rank_test_score
{'n_estimators': 400, 'min_samples_split': 5, 'min_samples_leaf': 2, 'max_features': 'sqrt', 'max_depth': None}	0.9497353701810310	0.94826186323450	0.9487723096817	0.9489231810324060	0.000610943152685858	1
{'n_estimators': 300, 'min_samples_split': 2, 'min_samples_leaf': 4, 'max_features': 'sqrt', 'max_depth': 30}	0.9495399049738380	0.94820172009380	0.9481407972093	0.9486274740923260	0.000645665282024003	2
{'n_estimators': 300, 'min_samples_split': 5, 'min_samples_leaf': 2, 'max_features': None, 'max_depth': 20}	0.9489084019967520	0.94856257893790	0.9481859052431	0.9485522953925610	0.000295047683437948	3
{'n_estimators': 200, 'min_samples_split': 2, 'min_samples_leaf': 2, 'max_features': None, 'max_depth': 10}	0.9492241534852950	0.94767546761290	0.9482911573218	0.948396926140	0.000636656508647184	4
{'n_estimators': 400, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_features': 'sqrt', 'max_depth': 20}	0.948818187285740	0.94763036025740	0.9479603650744	0.9481363042058350	0.000500632397271236	5

Binomial Logistic Regression

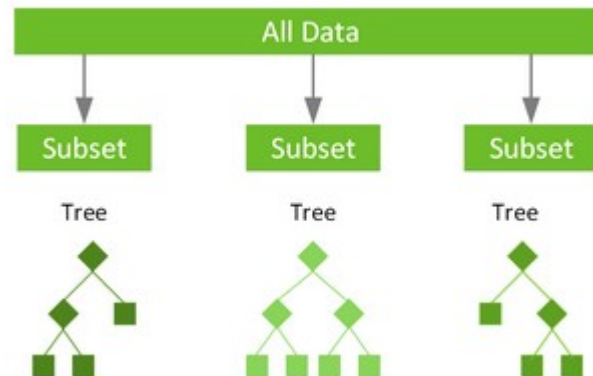
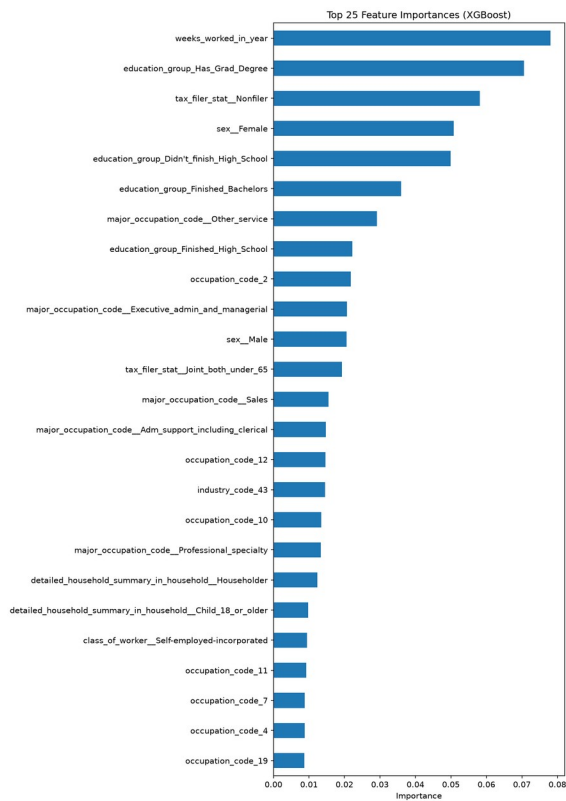


Logistic Regression accuracy: 0.9758

Classification report:

	precision	recall	f1-score	support
0	0.96	0.99	0.97	93576
1	0.69	0.34	0.46	6186
accuracy			0.95	99762
macro avg	0.82	0.67	0.72	99762
weighted avg	0.94	0.95	0.94	99762

XGBoost



XGBoost accuracy: 0.9510

Classification report:

	precision	recall	f1-score	support
0	0.96	0.99	0.97	93576
1	0.71	0.35	0.47	6186
accuracy			0.95	99762
macro avg	0.83	0.67	0.72	99762
weighted avg	0.94	0.95	0.94	99762

Outcome

- A model that is 95% accurate.
- Script that can retrain model, run hyper parameter search, or run the model on new data
- Key Features (minus anything income based)
 - Age
 - Education
 - Industry/Occupation

Running the script

```
### Train the model

```bash
python3 main.py --mode train
```

### Run the saved model on the test dataset

```bash
python3 main.py --mode run
```

### Tune hyperparameters

```bash
python3 main.py --mode tune --n-iter 20 --cv 3
```

### Custom files and label column

```bash
python3 main.py --mode train --train-file census_income_learn.csv --test-file census_income_test.csv --label-col label
```

## Column header utility

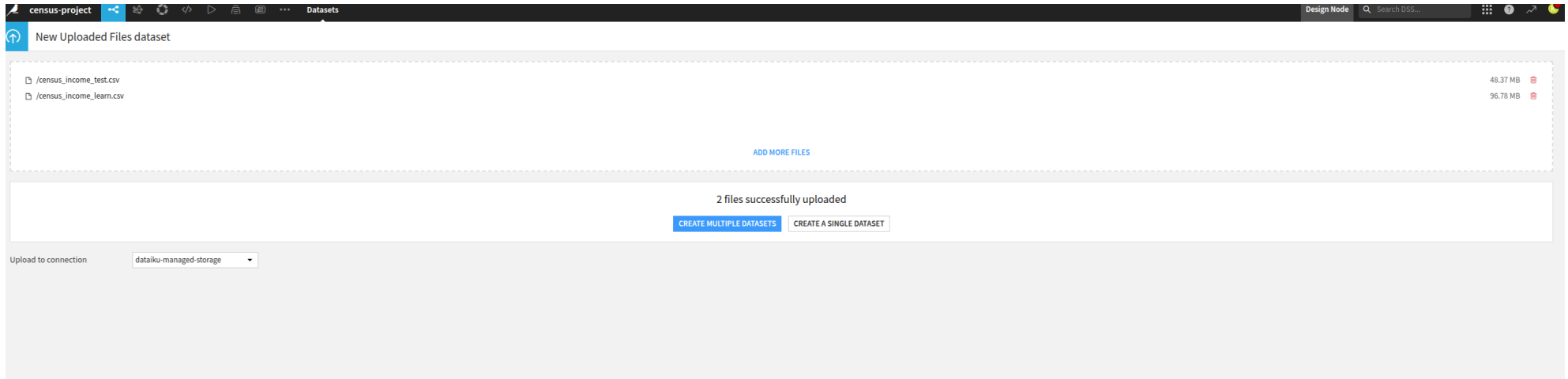
If your CSV files are missing headers, use `add_csv_headers.py` to prepend headers from the metadata file.

```bash
python3 add_csv_headers.py
```
```

Recommended Next Steps

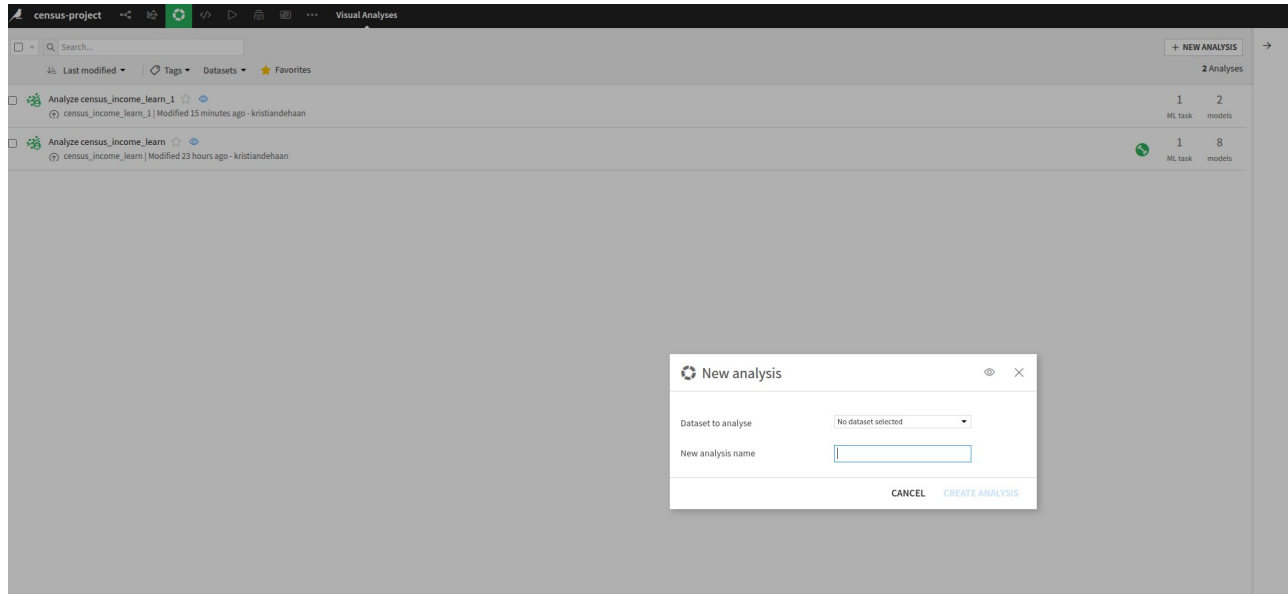
- Visualize the data on a GIS map to make it easy for social workers to find areas in needs.
- Run model on more the larger or newer census dataset
- Setup system to monitor for model drift and retraining as needed
- Setup API for 3rd party contractors to use the model
- Fix bug with adding headers to csv
- Improve Feature Engineering
- Try more models

How to do this yourself in Dataiku



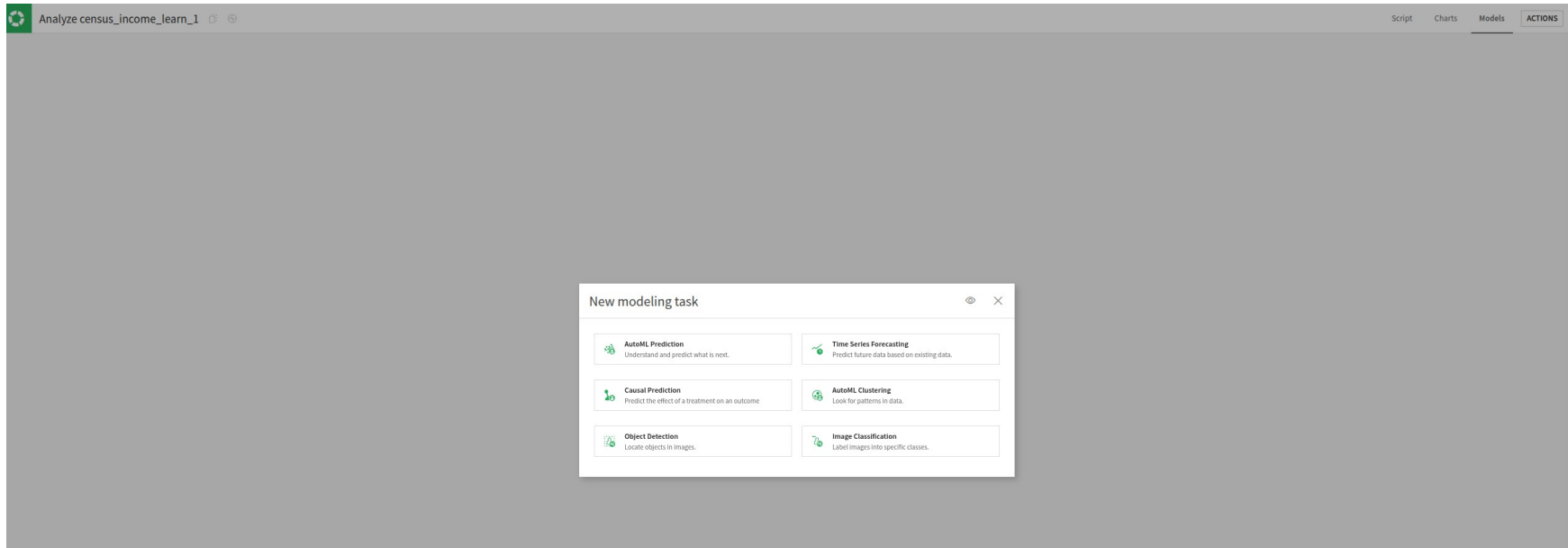
- Upload your data into a project, you can create two datasets or make one single dataset and re-split it later

How to do this yourself in Dataiku



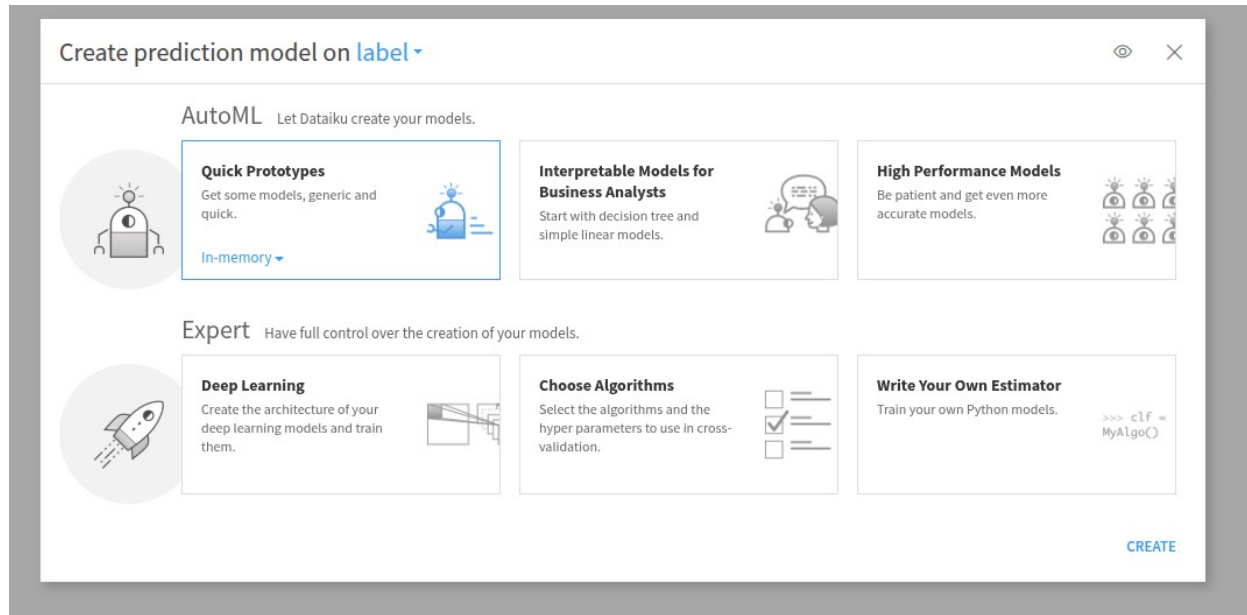
- Go to the Visual ML Tab and Create a new Analysis and select the learn dataset

How to do this yourself in Dataiku



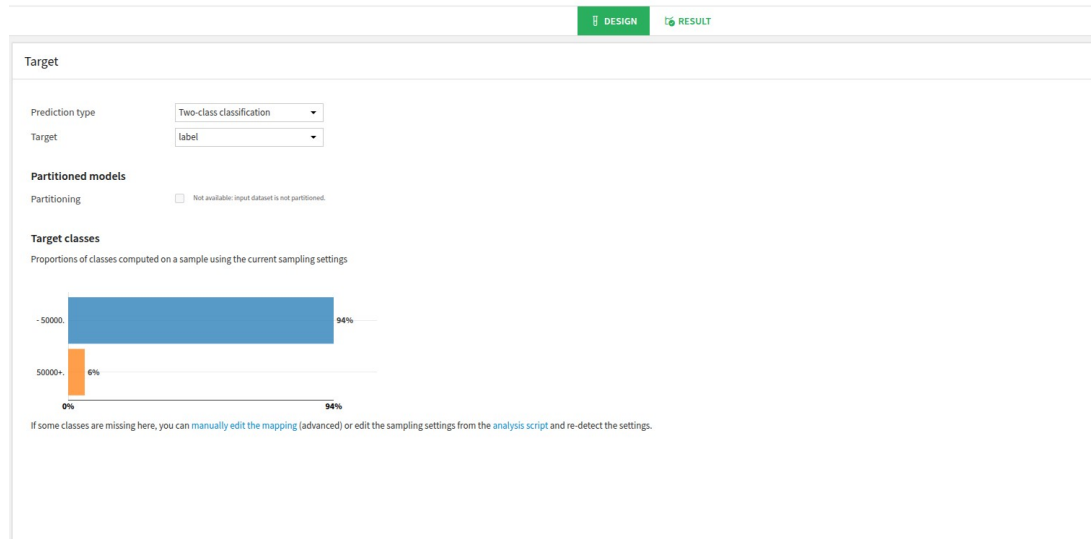
- Click it on the AutoML Prediction from the models tab

How to do this yourself in Dataiku



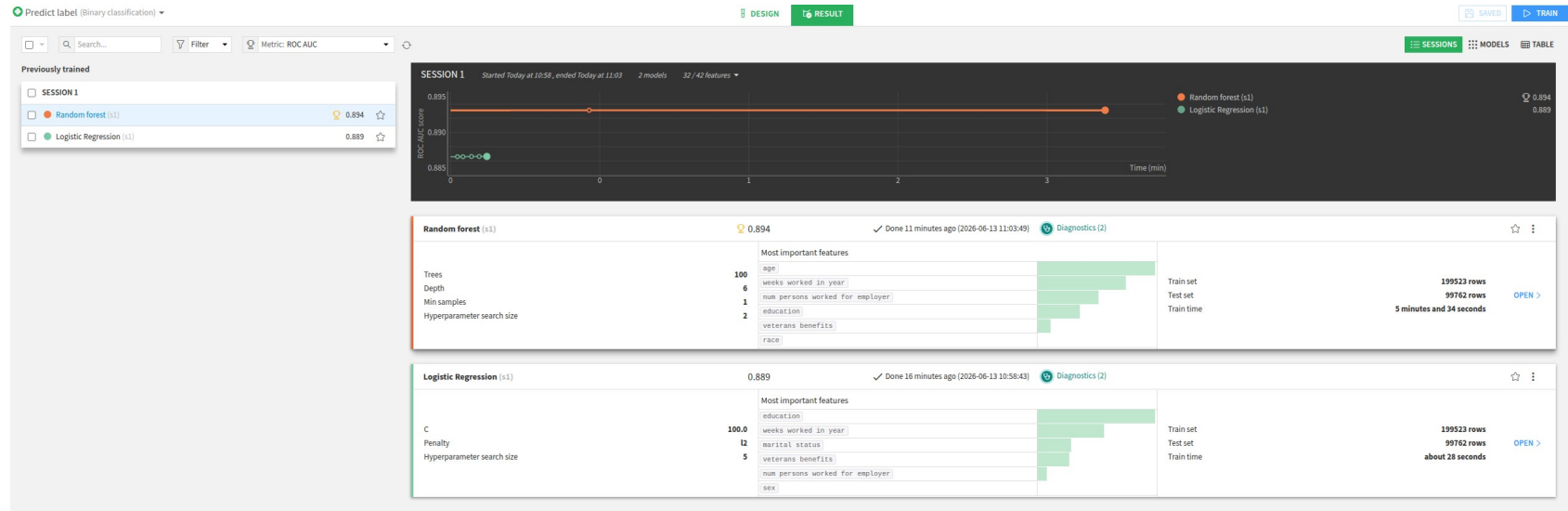
- Click the feature you want to predict and for now start with quick prototypes

How to do this yourself in Dataiku



- In the design tab we will set what features we want to use, what models we want to try and how to handle the train and test datasets.

How to do this yourself in Dataiku



- Click Train and let the model go, notice this model did not do as well as my coded model. This is just a first pass no data cleaning has happened outside what Dataiku will auto try.

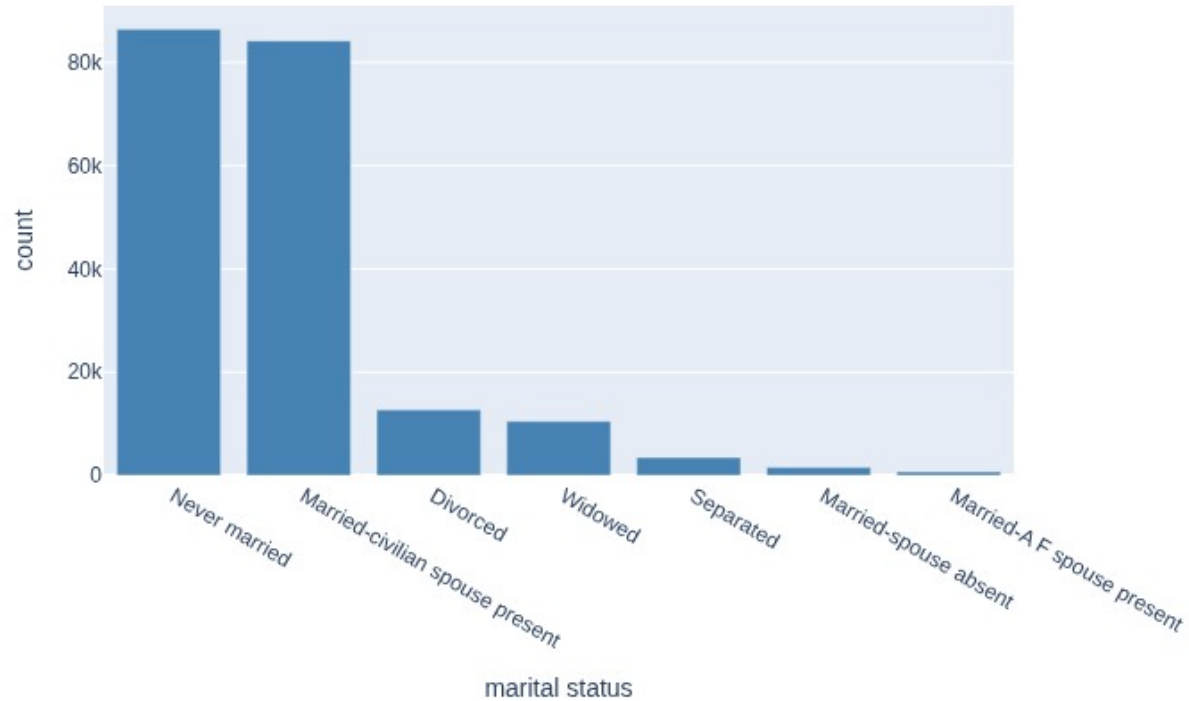
Questions?

Low Impact Features (backup)

- year
- capital losses
- migration code-change in reg
- migration prev res in sunbelt
- migration code-move within reg
- migration code-change in msa
- enrolled in edu inst last wk
- hispanic Origin
- wage per hour
- country of birth father
- member of a labor union
- race
- country of birth mother
- country of birth self
- region of previous residence
- citizenship
- reason for unemployment
- fill inc questionnaire for veteran's admin
- state of previous residence
- own business or self employed

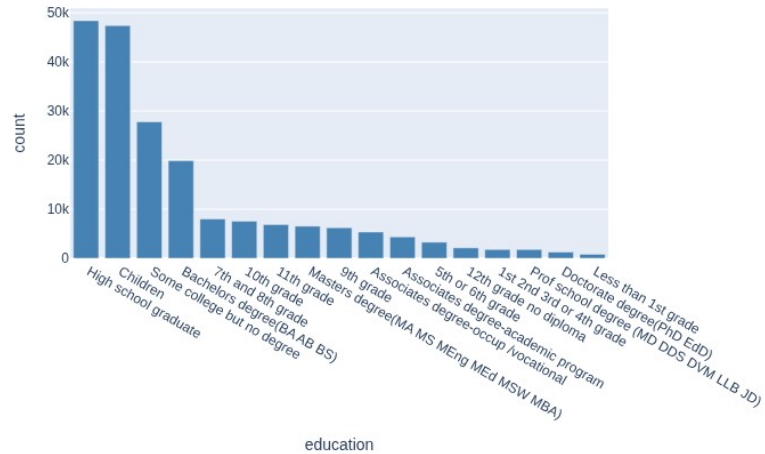
Marital Status

marital status value counts



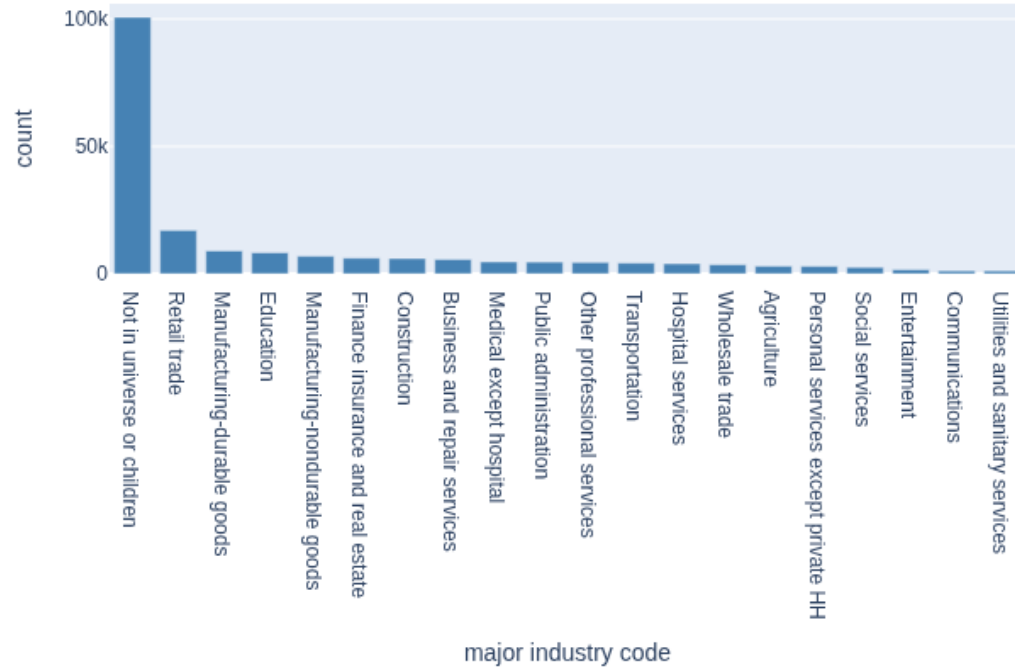
Education

education value counts



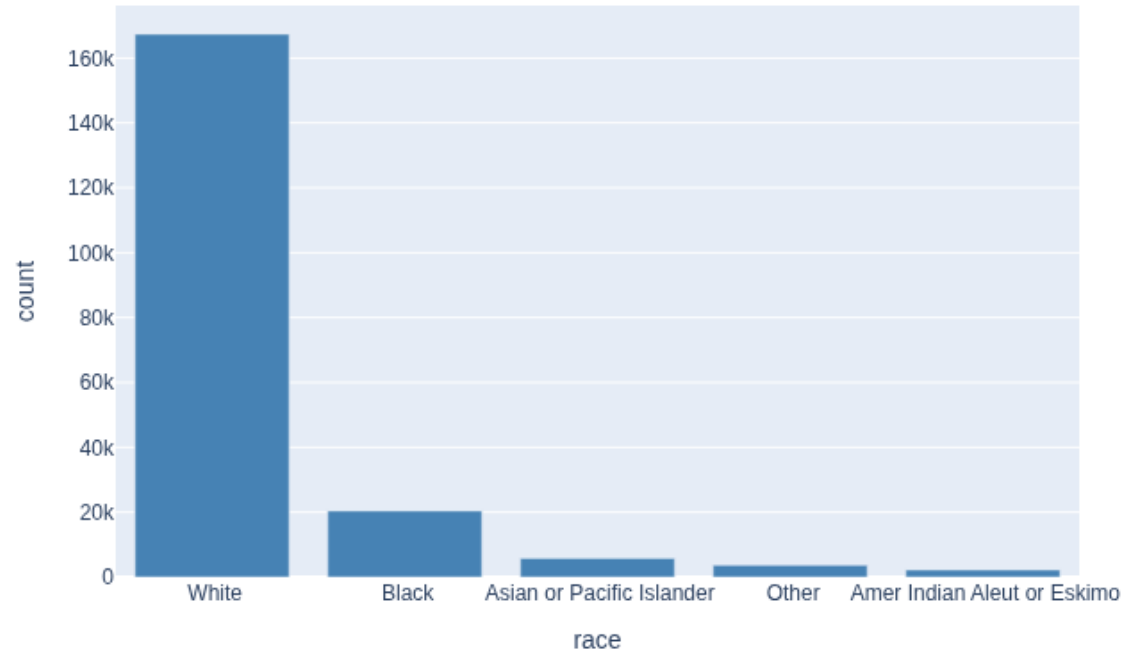
Major Industry

major industry code value counts



Race

race value counts



Marital Status

marital status value counts

